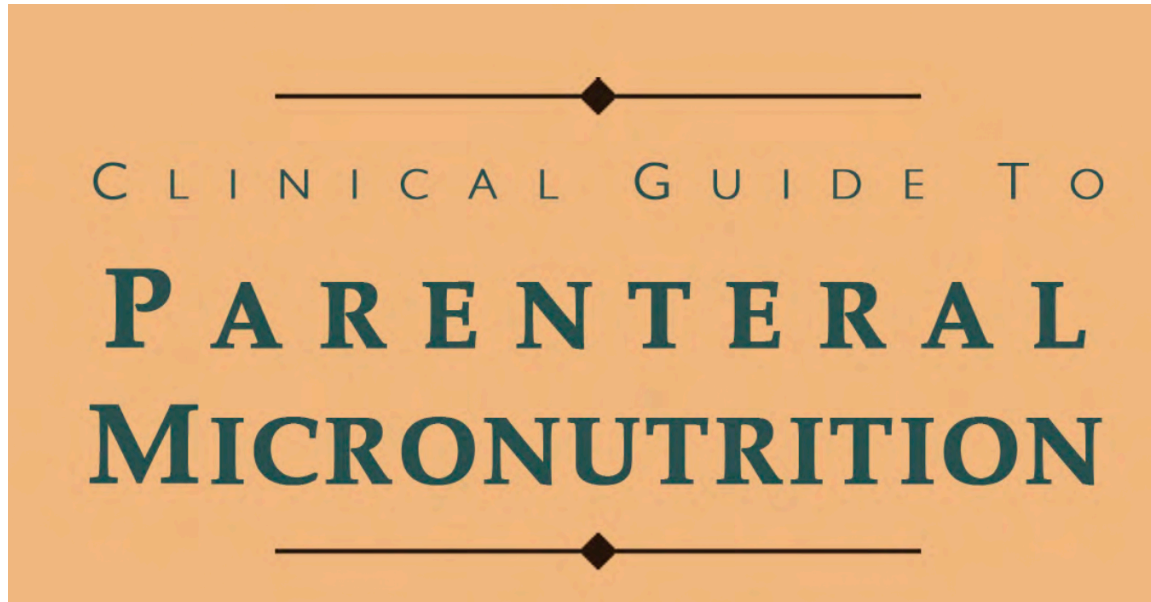




Self Assessment Free Primer

Electrolytes

by

Ed: Thomas G. Baumgartner, Pharm.D., M.Ed.

Baumgartner - Electrolytes Primer

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I. Electrolytes Q&A Self Assessment

SELF ASSESSMENT - CALCIUM

1. Calcium is mobilized from the bone by two hormones, _____ and _____ to maintain homeostasis of plasma concentrations of calcium.

parathyroid hormone (PTH, parathormone), vitamin D

2. The action of _____ is to antagonize parathormone and inhibit bone resorption by lowering both serum calcium and phosphate.

calcitonin

3. _____ (*mineral*) and _____ factor are intimately associated with at least one major route of B12 absorption from the distal ileum.

calcium and intrinsic

4. The mediation of a protein (using vitamin K) called _____ in hypercalcemic, hemodialyzed patients may cause bone calcium resorption, resulting in even more severe hypercalcemia.

osteocalcin

5. Calcium gluconate, in fact, can potentially contribute greater than 80% of the total _____ load from TPN and the accumulation of this element in bone may be the etiology of metabolic bone pain.

aluminum

6. _____ is the only pediatric amino acid formula that contains significant amounts of cysteine (about 15 mg/dL).

TrophAmine®

7. ____ appears to be the primary determinant of calcium/phosphate solubility.

pH

8. _____ increases serum calcium concentration by promoting osteoclastic activity and calcium resorption from bone, by increasing proximal renal tubular calcium resorption, and by enhancing gastrointestinal tract absorption of calcium through formation of 1,25-dihydroxycholecalciferol and Ca-binding protein.

Parathormone or parathyroid hormone

9. Most patients with hypocalcemia have a ____ (low,high) intracellular magnesium content.

low

10. ____% of oral calcium is absorbed from the small intestine (duodenum and

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proximal jejunum)

30

11. The RDA for pregnant (13 weeks or more) and lactating women is based on ___% bioavailability.

50

12. Once calcium gains entry into the systemic circulation it becomes ___% - ___% protein-bound by albumin.

40 - 50

13. _____ (hypercalcemia, hypocalcemia) presents as a shortened QT interval with an ST segment shift.

Hypercalcemia

14. Ionized calcium represents approximately ___% - ___% of plasma calcium while the remainder is primarily bound to albumin.

50-60

15. Approximately 15% of critically ill surgical patients develop _____, which may represent a new form of hyperparathyroidism.

hypercalcemia

16. The product of serum calcium and serum phosphate is used as a clinical guide in the assessment of soft tissue calcium / phosphate deposits (when the product exceeds ___ - ___ mg / dL) and defective mineralization of bone (when the product is below 30-40 mg / dL).

60-70

17. The circadian acrophase (highest point of the day) of total calcium, as well as actual (*ionized*) calcium, is approximately ___ pm, while circadian nadirs are at about 6:00 am and 2:00 am

12:00 Noon

18. Serum from both normal subjects and critically ill patients contained an ether-extractable factor which _____ (lowered, raised) ionized calcium concentrations and increased albumin-calcium binding.

lowered

19. Lipid infusions have _____ (lowered, raised) mean serum ionized calcium values in critically ill patients.

lowered

20. The best clinical evaluation of ionized calcium is achieved by taking measurements from _____ (sample fluid).

anerobically collected whole blood

21. Approximately ___% of filtered calcium is reabsorbed from the kidney, while less than 1% is excreted.

99

22. Increased _____ (about 20% greater than continuous) occurred with the

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use of cyclic TPN (75% of cyclic TPN patients were in negative calcium balance).

hypercalciuria

23. Hypercalciuria induced by high concentrations of amino acid infusion during TPN may contribute to the development of _____.
_____.

metabolic bone disease

24. Since cardiac function is intimately related to the calcium cation, intravenous calcium solutions should be administered at a rate no greater than ____-____ mEq/min.

0.75-3

25. Dietary phosphorus has been observed to have a _____ effect.

hypocalciuretic

26. TPN-induced hypercalciuria can be attenuated in the short term by intravenous _____.

phosphate

27. Metabolic bone disease is characterized by _____, _____, _____ and _____.

hypercalcemia, hypercalciuria, osteomalacia and bone pain

28. Calcium/phosphate salts deposit in soft tissues more frequently in the face of _____ and occur predictably at concentrations of serum calcium concentrations of 17 mg/dL or higher.

hypercalcemia

29. _____ calcium intake has been associated with hypertension.

Reduced

30. _____'s phenomenon is elicited by applying a blood pressure cuff to an upper extremity and inflating the cuff above the systolic blood pressure for three minutes.

Trousseau

31. _____'s sign identifies tetany as a unique lateral contraction of the facial muscles and the muscles of the eyelids elicited by percussion of the facial nerve anterior to the ear.

Chvostek

32. Cardiac standstill can occur at plasma calcium concentrations of greater than ____ mg/dL.

18

33. Elevated chloride to phosphate ratios beyond 33 have been associated with the _____ of primary hyperparathyroidism.

hypercalcemia

34. _____ (acidemia, alkalemia) causes calciuria.

acidemia

35. Although more chemically reactive, replacement of calcium gluconate by

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calcium chloride may help to reduce the aluminum content by approximately ____ % in the final parenteral solution

34

36. Adding _____ (alkalinizers, acidifiers) to total parenteral nutrition solutions has been associated with less urinary calcium excretion.

alkalinizers

37. In the setting of dialysis and _____ (hypercalcemia, hypocalcemia), vitamin K which mediates osteocalcin and promotes resorption should be avoided.

hypercalcemia

38. Patients at risk for hypervitaminosis A (vitamin A is osteolytic), such as acute and chronic renal failure patients, may promote _____ (hypercalcemia, hypocalcemia).

hypercalcemia

SELF ASSESSMENT - MAGNESIUM

1. Approximately ____% of the total magnesium in the body is present in soft tissue and ____% is in bone.

50, 50

2. At least ____mEq of magnesium/gram of nitrogen are needed for anabolism.

2

3. Parenterally administered magnesium is a reliable and effective treatment to prevent and control convulsions associated with ____ and ____.

preeclampsia and eclampsia

4. Adequate ____ (macronutrient) intake is necessary for optimum total body magnesium retention.

protein

5. High intakes of ____ (energy substrate) appear to increase the need for magnesium.

glucose

6. Magnesium should be administered cautiously to patients with ____ impairment

renal

7. Digitalis intoxication may be caused by magnesium ____.

deficiency

8. About 30%-50% of magnesium absorption occurs in the ____

parenchyma.

jejunoileal

9. Magnesium is bound approximately 30% to plasma ____.

albumin

10. Many authorities consider that ____ ____ magnesium content most accurately reflects magnesium status.

skeletal muscle

11. Magnesium supplementation has been shown to decrease ____ excretion.

potassium

12. ____ hormone enhances tubular reabsorption of magnesium.

Parathyroid

13. ____ (macrosubstrate) loading may cause increased urinary magnesium loss.

Protein

14. Increased urinary magnesium (also calcium and phosphorus) has been observed in patients receiving ____ versus continuous TPN solution administration.

cyclic

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15. The maximum intravenous magnesium dosing should not exceed _____ mEq/minute.

12

16. An _____ (increase or decrease) in conduction time (with lengthened PR and QRS intervals), hypotension, premature ventricular contraction and cardiac arrest may all present with elevated plasma magnesium concentrations.

increase

17. The central nervous system manifestations of hypermagnesemia include decreased neurotransmitter release and CNS _____.

depression

18. Magnesium _____ (salt) has little associated diarrhea because it is a more soluble salt.

gluconate

19. The average daily parenteral nutrition requirement for magnesium is ____ - ____ mEq/day.

10-45

20. Magnesium intoxication is manifested by a sharp drop in _____ and respiratory paralysis.

blood pressure

21. A(n) _____ (excess, deficit) of magnesium can cause cardiac effects (ie, prolonged P-R and QRS intervals, elevated T waves, AV block or PVCs).

excess

22. Hypermagnesemia may be treated with 1-2 grams of _____.

calcium gluconate

23. _____ (Prolonged, Lack of) exercise may depress serum magnesium.

Prolonged

24. One gram of magnesium sulfate will provide ____ mEq magnesium.

8.1

25. Deep tendon reflexes disappear at magnesium concentrations of ____ mEq/L.

4

26. Magnesium produces peripheral _____ (vasodilation, vasoconstriction).

vasodilation

27. Magnesium is used as a tocolytic (T/F).

T

28. Magnesium is used in the treatment of _____ withdrawal.

alcohol

29. Magnesium is effective as a _____ (bronchodilator, bronchoconstrictor).

bronchodilator

30. _____ (electrolyte) can antagonize the effects of magnesium toxicity.

Calcium

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31. Intravenous magnesium is generally used for toxemia of pregnancy
_____ (prior to, during or after) delivery.

two hours prior to delivery

32. Magnesium _____ (decreases or increases) acetylcholine and blocks neuromuscular transmission to prevent or control convulsions.

decreases

SELF ASSESSMENT - PHOSPHATE

1. Phosphorus, as _____ (intracellular anion), is important as an extracellular buffer, component of phospholipids, constituent of nucleic acids in chromosomes and ribosomes and mediator of energy transport (in concert with the nucleotides ATP and GTP).

phosphate

2. Phosphate is found in erythrocytes as 2,3 - _____ and the maintenance of oxygen release from hemoglobin.

diphosphoglycerate

3. Two phosphate salts, _____ and _____, are used in parenteral therapy.

monobasic and dibasic

4. Phosphate's role in adenosine triphosphate and its action as 2,3-diphosphoglycerate (2,3-DPG) and the release of oxygen from hemoglobin thereby promoting a beneficial swing to the _____ in the oxygen:hemoglobin dissociation curve.

right

5. _____ and phosphate solubility presents a major problem to the pharmacist in compounding parenteral nutrition solutions.

Calcium

6. ____ plays a major role in solubility phenomena.

pH

7. There is a reciprocal relationship between serum _____ and phosphate.

calcium

8. Approximately 70% of all phosphate is absorbed, primarily from the _____ and may increase to 90% in dietary phosphate restricted states.

jejunum

9. Serum _____ enzymes may increase dramatically when patients are transitioned from parenteral nutrition or a period of bowel rest to enteral nutrition.

alkaline phosphatases

10. In the extracellular fluid, phosphate is present primarily in the ionic form, however in bone it is available as _____.

phosphorus

11. The _____ phosphate parenteral salt is the more soluble.

monosodium

12. The buffer effects of phosphate are more pronounced in the _____ than in _____ fluids.

intracellular, extracellular

13. A decreased serum phosphate encourages more calcium to be resorbed into

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the blood from bone, while an increased phosphate plasma concentration stimulates the effect of _____, which facilitates deposition of calcium into the bone.

calcitonin

14. Children and neonatals have _____ (higher, lower) serum phosphate concentrations than adults.

higher

15. Serum phosphorus may _____ (increase, decrease) during states of dehydration, after meals and transiently after exercise.

increase

16. Hypophosphatemia symptoms ____ (may or may not) appear, even when serum phosphorus concentrations fall below 1 mg/dL.

may not

17. Since serum phosphate concentrations may vary as much as ____ - ____ mg/dL/day and increase after meals, the serum phosphate should be obtained at the same time in the morning before breakfast.

1-2

18. Phosphate homeostasis is regulated in the bone, gut and kidney by _____ hormone (which increases renal phosphate excretion and increases intestinal absorption of phosphate), growth hormone and vitamin D (both of which increase phosphate renal reabsorption and intestinal absorption).

parathyroid

19. About _____% of the approximately 700-800 g of phosphate in the human body is in the bone.

85

20. _____ (increased, decreased) urinary losses of calcium, phosphorus and magnesium occur in patients receiving cyclic versus continuous TPN.

Increased

21. Hemodialysis (HD), in patients with mild elevations of phosphorus receiving HD three times a week, removes only ____ mM of phosphate per run while peritoneal dialysis (PD) removes ____ mM/day of phosphate.

8, 10

22. Exceeding a serum calcium phosphate solubility product of ____ - ____ mg^2/dL^2 , usually in the face of hypercalcemia, may produce calcium phosphate deposition in soft tissue.

60-70

23. ____ (hypo, hyper) phosphatemia has been associated with significant reduction of neutrophil phagocytosis, intracellular killing, consumption of oxygen and generation of superoxide during phagocytosis.

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Hypo

24. ____ (hypo, hyper) phosphatemia may decrease high-energy substrate availability at the cellular level leading to respiratory muscle dysfunction.

Hypo

25. Intravenous glucose, antacids, diuretics and steroids are the most common agents associated with profound ____ (hypo,hyper) phosphatemia.

hypo

26. Acute ____ (hypo, hyper) phosphatemia can result in death.

hypo

27. The degree of fatty liver infiltration [since elevations of SGOT enzymes have correlated significantly with ____ (low, high) serum phosphate concentrations] may increase with (hypo, hyper) phosphatemia.

low, hypo

28. One millimole of phosphorus is equal to ____ mg.

31

29. Lipid emulsions contain approximately ____ mM phosphate per 500 mL.

7.5

30. Hyperphosphatemia per se does not have any adverse clinical effects except for ____.

ectopic calcifications

31. There are approximately ____-____ mg/day of phosphorus in the stool while urinary excretion amounts to approximately 700-800 mg/day.

100-200

32. Chronic hypophosphatemia may induce ____ in children and ____ in adults.

rickets, osteomalacia

33. A dosage of ____ mM/kg over 6 hours has been suggested for recent uncomplicated hypophosphatemia, while ____ mM/kg over 6 hours are recommended for prolonged, multiple-etiology hypophosphatemia.

0.08, 0.16

34. Beta blockers can block the ____ (hypophosphaturia, hyperphosphaturia) of acute respiratory alkalosis in humans.

hypophosphaturia

35. Chronic intravenous phosphate administration generates and maintains renal metabolic alkalosis in salt-replete humans and induces ____.

hyperparathyroidism

36. Sodium ____ (phosphate ester) has been shown to be of value in the treatment of neonatal hypophosphatemia.

glycerophosphate

SELF ASSESSMENT - BICARBONATE

1. Anions that are metabolic precursors to bicarbonate are _____, _____, _____ and _____.

acetate, citrate, gluconate, lactate

2. The normal ratio between bicarbonate and carbonic acid at pH 7.4 is ____:____.
20:1

3. A renally compromised patient having a pH of 7.38, base excess of -7.6, chloride of 111 and bicarbonate of 17, is metabolically _____ (alkalemic, acidemic).

acidemic

4. Carbon dioxide is considered to represent _____ (acid, base). status
acid

5. Bicarbonate is the principal _____ (extracellular, intracellular) buffer in the body

extracellular

6. Citrate (e.g., Shohl's, Poly Citra, Bi Citra) is frequently contaminated with high concentrations of _____ (trace element).

aluminum

7. Calculate the amount of bicarbonate (Body weight = 70 Kg and bicarbonate = 24-28 mEq/L) needed in a patient with a serum pH of 7.47, base excess of +8.2, potassium of 2.1 mEq/L and a serum bicarbonate of 40 mEq/L.

None, since the patient is clearly alkalemic; fatal hypokalemia may follow additional alkalinization.

8. Calculate the amount of bicarbonate (body weight = 70 Kg and bicarbonate = 24-28 mEq/L) needed in a patient with a serum pH of 7.2, potassium of 6.5 mEq/L and a serum bicarbonate of 18 mEq/L

$0.5 \text{ (L/Kg)} \times \text{Body Weight (Kg)} \times (\text{desired bicarbonate} - \text{mEq/L of bicarbonate}) = \text{dose}$
 $0.5 (70 \text{ kg}) \times (26 \text{ minus } 18) = 280 \text{ mEq}$

9. Approximately _____ of calculated bicarbonate deficit should be initially administered.

1/2 (then recheck blood gases)

10. Lactate to bicarbonate metabolic conversion requires ____ to ____ hours.
one, two

11. Extracellular fluid bicarbonate concentration is ____-____ mEq/L and dissolved carbon dioxide represents ____-____ mEq/L.

23-28; 1-2

12. Intracellular bicarbonate is typically around ____ mEq/L.
10

13. Acetate and gluconate are also converted to bicarbonate *in vivo* on an

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equimolar basis, with their metabolism apparently occurring _____ (intra-, extra-) hepatically.

extra

14. In the presence of carbonic anhydrase, the combination of carbon dioxide and water produces carbonic acid that dissociates into _____ (which is reabsorbed) and the _____ ion (which is actively secreted).

bicarbonate; hydrogen

15. An _____ (excess, deficit) of bicarbonate decreases oxygen release from hemoglobin to tissues.

excess

16. Approximately __ units of whole blood or __ units of fresh frozen plasma can produce clinically significant alkalemia.

8, 10

17. When acidosis or alkalosis is present, which electrolyte should be carefully monitored due to changes in distribution pattern?

potassium

18. Bicarbonate administration may _____ (decrease, increase) serum potassium.

decrease

19. Patients with chronic respiratory diseases who retain carbon dioxide will have _____ (decreased, increased) bicarbonate (frequently designated as serum CO₂)

increased

20. TPN acetate content may exacerbate acid / base balance in patients with pulmonary dysfunction (true, false).

true (acetate can alkalinize and precipitate respiratory carbon dioxide compensation)

SELF ASSESSMENT - CHLORIDE

1. Chloride is the principal anion of the _____ (*intracellular, extracellular*) fluid. **extracellular**

2. The _____ (*intracellular, extracellular*) chloride concentration is only about 1 mEq/L. **intracellular**

3. The major indication for chloride is in metabolic alkalemia where urine chloride concentration is _____ (*less, greater*) than 10 mEq/L accompanied by extracellular fluid volume depletion and potassium loss is characteristic.

less

4. With the exception of ammonium chloride, which is acidic, the chloride salts generally produce _____ (*acidic, basic, neutral*) aqueous solutions since they are derived from strong acids (*hydrochloric acid*) and bases (*sodium hydroxide, potassium hydroxide*).

neutral

5. Metabolically, chloride ions will, in turn, _____ (*increase, decrease*) serum hydronium ions.

increase

6. Acetate ions have been suggested to _____ (*promote, compete for*) renal tubular reabsorption of chloride.

compete for

7. Intracellular chloride also plays a unique role in the red blood cell by being exchangeable for _____ (*hydronium ion, bicarbonates*) in the plasma.

bicarbonates

8. Since chloride is excreted with _____ (*potassium, sodium*), many of the cation features regarding excretion also apply to chloride.

sodium

9. The kidney, sweat and particularly, _____ fluids represent the major routes of chloride loss in the body.

gastrointestinal

10. Like sodium, chloride is completely filtered by the glomerulus; over ____ (25, 50, 99) % is then reabsorbed along the proximal tubule, loop of Henle, distal tubule and collecting duct.

99

11. The majority of chloride reabsorption occurs via a(n) _____ (*active, passive*) transport process in the ascending limb of the loop of Henle.

active

12. Urine chloride concentration should increase from ____ to ____ mEq/L as normal acid-base balance returns.

60-70

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13. ____ teaspoonfuls of sodium chloride per liter will provide normal saline.

Two

14. What is the chloride deficit in a 70 kg patient having a chloride level of 85 mEq/dL (*nl=100mEq/dL*)? **Chloride deficit (mEq) = 0.2 L/kg x Wt (kg) x (normal chloride – measured chloride) = 210**

mEq chloride

15. In metabolic alkalemia, urinary chloride is ____ (*less, greater*) than 10 mEq/L.
less

16. Sodium and chloride in TPN are usually added in similar concentrations, however, severe, uncompensated metabolic _____ (acidemia, alkalemia) would require the use of a salt other than chloride.

acidemia

17. _____ alkalemia is often characterized by a hypernatremia in the setting of a high pH and positive base excess.

Contraction

18. What calcium salt should be avoided, although it has relatively less aluminum contamination, in TPN solutions?

Chloride

SELF ASSESSMENT - POTASSIUM

1. Potassium is the principal _____ (intracellular, extracellular) cation in the body.

intracellular

2. A typical 70 kg male has a total of approximately 3,500 mEq of potassium of which ____ % is intracellular.

98

3. Since total body stores remain within a narrow range, daily loss balances daily intake with about ____ % eliminated renally and the remainder eliminated via the sweat and feces.

90

4. The kidneys can adjust quickly to increased intake and ____ (can, cannot) prevent depletion in the absence of any potassium ingestion.

cannot

5. Increased potassium renal loss can occur in the presence of high concentrations of nonreabsorbable anions (e.g., penicillins) that _____ (decrease, increase) the intraluminal negativity of the distal tubule and promote potassium excretion.

increase

6. Adrenal insufficiency produces potassium _____ (excretion, retention) and sodium _____ (retention, wasting).

retention, wasting

7. Insulin promotes an _____ (intracellular, extracellular) potassium shift.

intracellular

8. Exercise results in potassium cellular _____ (influx, efflux) and venous blood concentrations _____ (decrease, increase) immediately.

efflux, increase

9. Only about ____ % of total body potassium is present in the extracellular fluid.

10

10. A ____ pH unit shift in the plasma results in 0.4-0.6 mEq/L potassium shift.

0.1

11. Approximately ____ % - ____ % of potassium is bound to plasma proteins.

10-20

12. During nutritional replacement, potassium is deposited in newly synthesized cells and up to ____ mEq/g to ____ mEq/g of administered nitrogen may be required.

5-6

13. In a 70 kg male, a decrease of ____ mEq/L in serum potassium from 5 mEq/L represents a deficit of 100-200 mEq.

1

14. Urinary potassium of less than ____ mEq/L is indicative of a deficit.

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40

15. Metabolic acidemia leads to _____ serum potassium.

increased

16. Angiotensin-converting enzyme inhibitor therapy may (increase, decrease) serum potassium.

increase

17. _____ (anticoagulant) has been associated with hyperkalemia.

Heparin

SELF ASSESSMENT - SODIUM

1. Sodium is the principal _____ (extracellular, intracellular) cation in the body.

extracellular

2. Normal extracellular fluid sodium concentration is about 140 mEq/L, while the intracellular fluids contain about ____ mEq/L.

10

3. Mineralocorticoid hormones act on the _____ (distal, proximal) tubule to enhance sodium reabsorption and potassium excretion.

distal

4. The kidney, sweat and _____ fluids represent the major routes of sodium excretion in the body.

gastrointestinal

5. At a normal glomerular filtration rate of 100 mL/min and a normal serum sodium concentration of 140 mEq/L, ____ mEq/min of sodium are filtered.

14

6. Patient PJ (70 kg) is dehydrated and has central nervous system manifestations of hyponatremia and a serum sodium of 120 mEq/L. Sodium replacement is _____ mEq.

208

[125 - measured serum sodium (mEq/L)] x 0.6 L/kg x wt. (kg) = required sodium (mEq), where 0.6 L/kg reflects normal total body water function and 0.6 x wt (kg) provides an estimate of total body water.

7. The amount of normal saline that would be required in previous patient PJ would be ____ L.

1.35

8. Hypertonic saline solutions (commercially available) are ____% - ____% sodium chloride.

3 -5

9. Acute hyponatremia is classified as any drop in serum sodium that exceeds the rate of ____ mEq/L/hr.

0.5

10. Continuous nasogastric suction combined with histamine₂-antagonists can result in large sodium losses due to the suppression of _____ ion resulting in rising gastric sodium content.

hydrogen

11. A 2 L/day loss by nasogastric tube at a gastric pH 4 can result in sodium losses of ____ mEq/day.

150

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12. A diabetes insipidus patient should receive _____ (increased, decreased) sodium in TPN.

decreased

13. Demeclocycline therapy will cause serum sodium to _____ (increase, decrease).

increase

14. Vasopressin therapy will _____ (increase, decrease) serum sodium.

decrease

15. Patients receiving chronic _____ therapy (manic depression medication) should receive meticulous serum sodium monitoring.

lithium

16. The fractional excretion of sodium greater than 1 indicates _____.

nephrosis

17. Serum osmolality is primarily a function of serum _____.

sodium

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Calcium

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ZINC

COPPER

CHROMIUM

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